

Advanced (Habanero)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-1/2$
(B) $1/2$
(C) 2
(D) -2
(E) 1
- Q2.** Which of the following lines is perpendicular to $y = -3x - 2$?
(A) $y = 1/3x + 1$
(B) $y = -1/3x - 1$
(C) $y = 3x - 1$
(D) $y = -3x + 1$
(E) $y = 1/3x - 2$
- Q3.** If $y = mx + b$ is perpendicular to $y = 2x - 1$, what is the value of m ?
(A) $-1/2$
(B) $1/2$
(C) -2
(D) 2
(E) 1
- Q4.** Which equation represents a line perpendicular to $y = x - 2$?
(A) $y = -x - 1$
(B) $y = -x + 1$
(C) $y = x + 2$
(D) $y = -x + 2$
(E) $y = x - 1$
- Q5.** The equation of a line perpendicular to $y = -4x + 1$ is $y =$
(A) $1/4x + 2$
(B) $-1/4x - 2$
(C) $4x - 1$
(D) $-4x - 1$
(E) $1/4x - 1$
- Q6.** If the line $y = 2x + 1$ is perpendicular to the line $y = mx - 3$, what is the value of m ?
(A) $-1/2$
(B) $1/2$
(C) -2
(D) 2
(E) 1
- Q7.** What is the equation of a line that is perpendicular to $y = -2x + 3$ and passes through $(0, 0)$?
(A) $y = 1/2x$
(B) $y = -1/2x$
(C) $y = 2x$
(D) $y = -2x$
(E) $y = 1/2x + 1$
- Q8.** If the line $y = -3x + 1$ is perpendicular to $y = mx + 2$, what is the value of m ?
(A) $1/3$
(B) $-1/3$
(C) 3
(D) -3
(E) 1

Open Ended Questions (FRQ)

- Q9.** Find the equation of a line that is perpendicular to $y = 2x - 1$ and passes through $(1, 3)$.
- Q10.** Show that the lines $y = 2x + 1$ and $y = -1/2x - 2$ are perpendicular.

Chef's Tips

- Emphasize the importance of using the negative reciprocal of the slope to find perpendicular lines.
- Encourage students to use the point-slope form to write equations of perpendicular lines.
- Remind students to check their work by verifying that the product of the slopes of the two lines is -1 .